Milestone: Project Proposal & Justification

Project Title: BIST-Algo-Analyzer: Borsa Istanbul Data and News Analysis System

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1. Problem Statement

One of the biggest challenges individual investors face in financial markets is that market data (numerical data) and market news (textual and sentiment data) are scattered across fragmented platforms. Today, when an investor wants to assess the current status of a stock, they must check the stock's OHLCV (Open, High, Low, Close, Volume) data on one app (such as TradingView or a brokerage app) and track the latest news about that company on another app.

This situation disrupts the analytical decision-making process and creates information asymmetry, leading investors to make incorrect or delayed emotional decisions.

The core solution of this project: Thanks to this relational database project we will develop, investors will be able to view the latest news about the stocks they own or track through a single system. At the same time, they will also be able to analyze OHLCV-style pricing data—simple yet critically important for technical analysis—all in one place. When they log into the system for a specific stock, they will see both the company's financial volume and moving averages, as well as sentiment analyses of the latest KAP disclosures. This infrastructure will significantly simplify investors' lives, keep data tracking up-to-date on a single screen, and enable them to make rapid financial decisions through an algorithmic system.

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2. Justification (Why It Matters)

The significance of the project we have developed lies in its ability to organically integrate quantitative stock market data with qualitative news sentiment scores within a database-centric architecture. Thanks to our expanded dataset based on the BIST 100 index, we will be able to generate clear SQL queries in seconds to answer complex, multi-parameter questions.

Additionally, rather than limiting the project to just 3–5 stocks, we expanded our scope to include all (approximately 100) companies within the BIST100 index. For each of these companies, we have 1 year of historical price data (250 days) and carefully processed KAP news. This "Big Data" approach offers a scale befitting the academic rigor of a database design course, justifying the transformation of this application from a simple storage tool into a professional "Financial Decision Support System."

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3. Schema Design (ERD & Table Descriptions)

Our project consists of a total of 8 tables designed in accordance with full normalization rules (1NF, 2NF, 3NF). The architecture successfully addresses One-to-Many (1:N) and Many-to-Many (M:N) relationships. Below is the Entity-Relationship Diagram (ERD) of our database architecture and detailed lists of table functions:

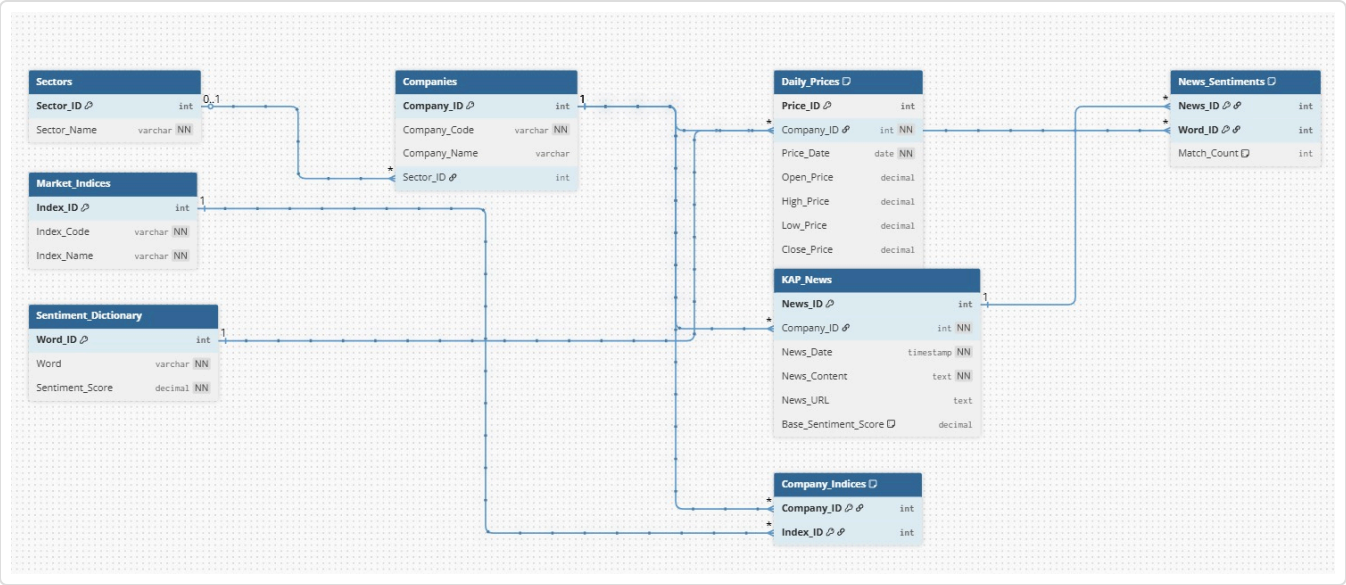

Table Descriptions

1. **Sectors:** This is the primary reference table containing the business sectors of BIST companies.
2. **Companies:** The main backbone table containing the codes (e.g., THYAO, ASELS) and basic company information.
3. **Market_Indices & Company_Indices:** A system linking market indices with companies.
4. **Daily_Prices:** This is where OHLCV data (Opening, Closing, Volume, etc.) is stored. It maintains detailed records of all trading days over a one-year period.

5. **KAP_News:** This is a news notification table that stores content in full format (as News_Content).
6. **Sentiment_Dictionary:** A financial word/score dictionary serving as the foundation for NLP analysis.
7. **News_Sentiments:** Our critical M:N junction table that enables news-word matching for algorithmic scoring.

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4. Proposed SQL Features

During the final project implementation phase, we actively used the following advanced SQL features to read data from the system and receive a perfect score evaluation under Category 3:

- **JOINs (INNER, LEFT):** To seamlessly link Price (Daily_Prices), Company (Companies), and News (KAP_News) data.
- **Aggregations (GROUP BY, SUM, AVG, COUNT):** Used when calculating total daily news sentiment score or average 10-day volume.
- **Views:** Used to store queries containing complex operations in the database as objects.
- **Window Functions (OVER, PARTITION BY):** Used to calculate the 50-Day Simple Moving Average (SMA-50).
- **CTEs (Common Table Expressions - WITH):** Used in a structured block to enhance the readability of our code.

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5. Sample Queries (Advanced Sample Queries)

Query 1: Analytical Opportunity Stocks View

```
 CREATE OR REPLACE VIEW Firsat_Hisseleri_View AS
SELECT c.Company_Code, dp.Price_Date, dp.Close_Price, dp.Volume,
       SUM(sd.Sentiment_Score * ns.Match_Count) AS Total_Daily_Sentiment_Score
FROM Companies c
JOIN Daily_Prices dp ON c.Company_ID = dp.Company_ID
JOIN KAP_News kn ON c.Company_ID = kn.Company_ID AND dp.Price_Date = DATE(kn.News_D
JOIN News_Sentiments ns ON kn.News_ID = ns.News_ID
JOIN Sentiment_Dictionary sd ON ns.Word_ID = sd.Word_ID
GROUP BY c.Company_Code, dp.Price_Date, dp.Close_Price, dp.Volume
```

```
HAVING SUM(sd.Sentiment_Score * ns.Match_Count) > 0
ORDER BY dp.Price_Date DESC;
```